

Meet a Deep-Learning Notebook

100 points • released September 10 • due September 14, 2026

In this assignment you will get comfortable with Google Colab, train a cat/dog classifier, inspect its predictions, and tour the other machine-learning examples in Chapter 1.

No prior Python experience is assumed. Your goal is to notice what goes into a model, what comes out, and what changes when you run an experiment.

START HERE

Open the course notebook

Open the guided course companion notebook.

1. Choose **File** → **Save a copy in Drive**.
2. Put your name at the top of your copy.
3. Write all answers in the notebook. This handout is a guide, not a separate response form.

Read alongside the original fast.ai Chapter 1 notebook.

New to notebooks?

Begin with the guided notebook. An optional 15-minute Colab quickstart is available if you want extra practice.

DURING CLASS

Our 60-minute plan

0–10 min	Save a copy, select a GPU if available, run and edit a cell, and complete Checkpoint 1.
10–25 min	Train the pet classifier. Find its inputs, labels, model, and training call. Begin Checkpoint 2.

25–30 min	Take a break while training finishes.
30–45 min	Predict first, then test the supplied images. Complete Checkpoint 3.
45–50 min	Break or catch up.
50–60 min	Discuss a result, write your own reflection, and review the homework.

The class goal is to reach the first classifier investigation, not to finish the chapter. If a download or runtime holds you up, ask for help and keep a note of the issue.

CHOOSE YOUR ROUTE

Two supported ways through

Core route • recommended

Use the course companion notebook. It has simplified setup, supplied validation images, and expensive examples switched off by default. This is a good choice if Colab or Python is new to you.

Challenge route

Use the original fast.ai Chapter 1 notebook. This is fully supported if you want to work closer to the textbook and troubleshoot its original workflow. Follow the alternate-route guide. The learning goals and grading are the same.

Full credit is available through either route. Clearly label results you ran yourself and textbook outputs you only inspected.

FINISH FOR HOMEWORK

Read, run, and compare

Read Chapter 1, including “What Is Machine Learning?”, the examples beyond image classification, and “Validation Sets and Test Sets.”

- Run the required pet, tabular, and movie-recommendation examples in your chosen notebook.
- For image segmentation and IMDB sentiment classification, inspect the code and saved outputs. Training these two examples is optional.

- Complete the five-example comparison table. Distinguish models you ran from results you inspected.
- You do not need to submit answers to the chapter's final questionnaire.

A simpler setup

- Select **Runtime** → **Change runtime type** → **T4 GPU** if available. No paid upgrade is required.
- The Core route does not mount Google Drive. Saving a notebook copy in Drive is a separate action.
- The required prediction cells use supplied images. Uploading an image is optional.
- Run cells in order. After a restart, rerun from the top.
- The Core route uses a smaller training subset and model, so your error rate will differ from the textbook's.

WHAT TO COMPLETE

100
POINTS

Six checkpoints

Checkpoint 1 · Getting oriented · 15 points

Name, saved copy, runtime, a changed calculation, and an explanation of how rerunning works.

Checkpoint 2 · What did we train? · 20 points

Inputs and labels, your validation error rate and its meaning, and the difference between training and prediction.

Checkpoint 3 · Inspect predictions · 25 points

Three image records: your prediction before running, the model output, and a brief interpretation. An uncertain or incorrect result is interesting; no particular score is required.

Checkpoint 4 · Human-only reflection · 15 points

Write 2–4 sentences about your learning, challenges, or next questions. Do not use AI to draft or polish this response.

Checkpoint 5 · Compare the examples · 15 points

For five examples, identify the dataset, input, output, model, and what you ran or inspected. Explain why evaluation data is held out.

Checkpoint 6 · Submission · 10 points

Complete answers, visible execution outputs, AI credit, and both required submission files.

SUBMIT IN CANVAS

Two files

1. Your completed `.ipynb` notebook: choose **File → Download**.
2. A PDF of that notebook with your own execution results: choose **File → Print → Save to PDF**.

Inspect the PDF before submitting. Your answers and relevant outputs should be visible and readable. No separate completed copy of this handout is required.

If a technical issue remains unresolved, keep the error and briefly describe what you tried. Do not present the textbook's saved outputs as your own.

AI and collaboration

You may use AI to understand code or debug. Credit the tool and what it helped with in the notebook. If you did not use AI, write "None."

You may discuss ideas and help classmates with setup, but submit your own executed notebook and explanations. Write the human-only reflection entirely yourself.

Optional preparation for Lab 2

- Install VS Code.
- Create a GitHub account and explore GitHub Education with your Yale email.

This preparation is not part of the Homework 1 grade.